

sed

The Ultimate Stream Editor Cheat Sheet

</> Basic Syntax

```
sed [OPTIONS] '[address]command[arguments]' [input_file ... ]
```

Reads text line-by-line, applies a script of commands, and prints the result to standard output.

🔧 Common Options

-n: Suppress automatic printing.

-i[SUFFIX]: Edit files in-place.

-E, -r: Use Extended Regex (ERE).

-e SCRIPT: Add a script to execute.

-f FILE: Read script from a file.

-s: Treat files as separate streams.

🎯 Addresses (Selecting Lines)

By line number
5, 10,20, 5,\$

By regex pattern
/pattern/

By range
/start/,/end/

Every Nth line
1~2 (every other)

Invert match
/pattern/! d (delete lines NOT matching)

📄 Core Editing Commands

s/old/new/flags: Substitute

d: Delete line

p: Print line (use with **-n**)

a \text: Append text after line

i \text: Insert text before line

c \text: Change/replace line

y/abc/xyz/: Transliterate chars

↔️ Hold Space & Flow Control

`sed` uses two buffers: the active **Pattern Space** and a storage **Hold Space**. These commands move data between them and alter the normal line-by-line flow.

h : Copy Pattern → Hold (overwrite)
H : Append Pattern → Hold

```
sed '1h; 2,$H; $g'  
# Collect all lines in hold space, then print at the end.
```

g : Copy Hold → Pattern (overwrite)
G : Append Hold → Pattern

```
sed 'G' file.txt  
# Double-space a file by appending hold space to each line.
```

x : Exchange Pattern ↔ Hold

```
sed -n '/A/{h;n;p;g;p}'  
# Swap line with 'A' and the line after it.
```

n : Print pattern, then read next line
N : Append next line to pattern (with \n)

```
sed '/START/{N; s/START\\nEND/MATCH/}'  
# Join lines to find a multi-line match.
```

q : Quit script immediately
= : Print current line number

```
sed '100q' bigfile.log  
# Process only the first 100 lines.
```

r file : Read and insert file content
w file : Write pattern space to file

```
sed '/ERROR/w errors.txt' logfile  
# Write all error lines to a separate file.
```

💡 Practical Examples

```
# Replace all 'foo' with 'bar' in-place with a backup
sed -i.bak 's/foo/bar/g' file.txt
```

```
# Print only lines between two patterns
sed -n '/START/,/END/p' logfile.log
```

```
# Swap first and last names (using backreferences)
sed -E 's/([^\ ]+) ([^\ ]+)/\2, \1/' names.txt
```

```
# Reverse every two lines using hold space
sed -n 'h;n;p;g;p' file.txt
```